

Homework 8

8.1 Judge whether the following statements are correct, and use "√" and "X" to indicate the judgment results.

(1) In the block diagram of figure 8.1, if A and F are the same symbols, there may be a sine wave oscillation.

(2) when the RC series parallel frequency selection network as feedback network, $\phi_F=0$, Single tube co amplifier circuit's $\phi=0$, $\varphi_A + \varphi_F = 2n\pi$ So they can form a reasonable connection of sine wave oscillation circuit

(3) As long as $|AF|=1$, there will be a sine wave oscillation

(4) Self excited oscillation can not be produced by negative feedback amplification circuit

(5) In the LC sinusoidal oscillating circuit, the reason that universal integrated OP amplifier is not used as a magnifying circuit is the upper cut-off frequency is too low.

(6) As long as the integrated operational amplifier is introduced into the positive feedback, it will work in the nonlinear zone.

8.1 判断下列说法是否正确，用“√”、“×”表示判断结果。

(1) 在图 P8.1 所示方框图中，只要 $\dot{A}$ 和 $\dot{F}$ 同符号，就有可能产生正弦波振荡。()

(2) 因为 RC 串并联选频网络作为反馈网络时的 $\varphi_F = 0^\circ$ ，单管共集放大电路的 $\varphi_A = 0^\circ$ ，满足正弦波振荡的相位条件 $\varphi_A + \varphi_F = 2n\pi$ (n 为整数)，故合理连接它们可以构成正弦波振荡电路。()

(3) 电路只要满足 $|\dot{A}\dot{F}| = 1$ ，就一定会产生正弦波振荡。()

(4) 负反馈放大电路不可能产生自激振荡。()

(5) 在 LC 正弦波振荡电路中，不用通用型集成运放作放大电路的原因是其上限截止频率太低。()

(6) 只要集成运放引入正反馈，就一定工作在非线性区。()

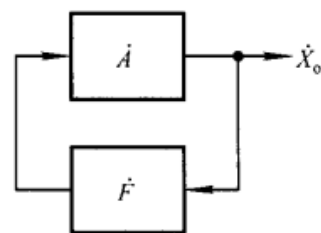


图 P8.1

8.2 Judge whether the following statements are correct, and use "√" and "X" to indicate the judgment results.

(1) in order to make the output voltage of voltage comparator be either high level or low level, it is necessary to make the integrated operational amplifier work in the open-loop state or just cause positive and negative results. ()

- (2) if the two threshold voltages of a pullback comparator are the same as those of a window comparator, their output voltage waveforms are the same when their input voltages are the same. ()
- (3) in the process of monotonous change of input voltage, the output voltage of single limit comparator and hysteresis comparator can only jump. ()
- (4) compared with hysteresis comparator, the single limit comparator has stronger anti-interference ability, while the hysteresis comparator is more sensitive than the single limit comparator.()

8.2 判断下列说法是否正确，用“√”、“×”表示判断结果。

- (1) 为使电压比较器的输出电压不是高电平就是低电平，就应在其电路中使集成运放不是工作在开环状态，就是仅仅引入正反馈。()
- (2) 如果一个滞回比较器的两个阈值电压和一个窗口比较器的相同，那么当它们的输入电压相同时，它们的输出电压波形也相同。()
- (3) 输入电压在单调变化的过程中，单限比较器和滞回比较器的输出电压均只可能跃变一次。()
- (4) 单限比较器比滞回比较器抗干扰能力强，而滞回比较器比单限比较器灵敏度高。()

8.3 Choose the following answer in the space

A.Capacitive B.Resistance C.inductive

(1) LC parallel network at resonant time_____;

When the frequency of the signal is larger than the resonant frequency_____;

When the frequency of the signal is smaller than the resonant frequency_____;

(2) When the frequency of the signal equals the series resonant frequency of the quartz crystal or the parallel resonant frequency, the quartz crystal_____;

When the frequency of the signal is between the series resonant frequency of the quartz crystal and the parallel resonant frequency, the quartz crystal_____;

Quartz crystal in the rest of the case _____;

(3) When the signal frequency is $f=f_0$, the RC series parallel network_____.

8.3 选择下面一个答案填入空内，只需填入 A、B 或 C。

A. 容性 B. 阻性 C. 感性

(1) LC 并联网络在谐振时呈 _____，在信号频率大于谐振频率时呈 _____，在信号频率小于谐振频率时呈 _____。

(2) 当信号频率等于石英晶体的串联谐振频率或并联谐振频率时，石英晶体呈 _____；当信号频率在石英晶体的串联谐振频率和并联谐振频率之间时，石英晶体呈 _____；其余情况下石英晶体呈 _____。

(3) 当信号频率 $f = f_0$ 时， RC 串并联网络呈 _____。

8.7 The circuit is as shown in figure p8.7. The stabilizing tube D_2 plays the role of stabilizing amplitude, and its stable voltage $U_2 = \pm 6V$. Try to estimate :

(1) the effective value of the output voltage without distortion;

(2) the oscillation frequency.

8.7 电路如图 P8.7 所示，稳压管 D_z 起稳幅作用，其稳定电压 $\pm U_z = \pm 6V$ 。试估算：

(1) 输出电压不失真情况下的有效值；

(2) 振荡频率。

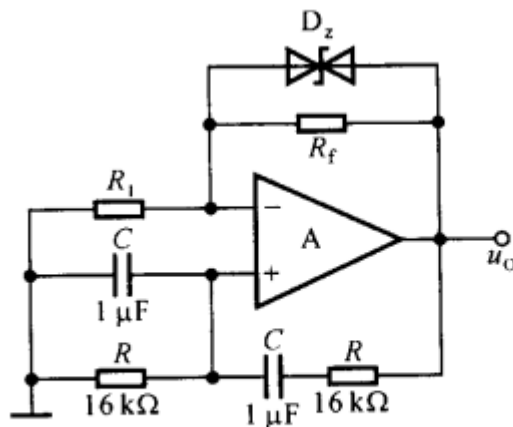


图 P8.7

8.8 The circuit is as shown in figure p8.8.

(1) In order to produce the sinusoidal oscillation of the circuit, mark the "+" "-" of the integrated OP amplifier and What kind of sinusoidal oscillating circuit is ?

(2) If R_1 is short circuited, what will the circuit produce?

(3) If R_1 is broken, what will the circuit produce?

(4) If R_f is short circuited, what will the circuit produce?

(5) If R_f is broken, what will the circuit produce?

8.8 电路如图 P8.8 所示。

(1) 为使电路产生正弦波振荡，标出集成运放的“+”和“-”；并说明电路是哪种正弦波振荡电路。

(2) 若 R_1 短路，则电路将产生什么现象？

(3) 若 R_1 断路，则电路将产生什么现象？

(4) 若 R_f 短路，则电路将产生什么现象？

(5) 若 R_f 断路，则电路将产生什么现象？

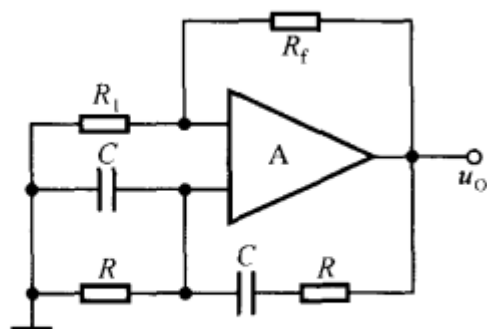


图 P8.8

8.9 The circuit shown in Figure P8.9 is a quadrature sine wave oscillator circuit, which can generate sine and cosine signals with the same frequency. Knowing that the stable voltage of the Zener tube is $\pm U_Z = \pm 6V$, $R_1 = R_2 = R_3 = R_4 = R_5 = R$, $C_1 = C_2 = C$.

(1) Try to analyze why the circuit can meet the conditions for sine wave oscillation;

(2) Find the oscillation frequency of the circuit;

(3) Draw the oscillograms of u_{O1} and u_{O2} . It is required to show their phase relationship and find their peak values respectively.

8.9 图 P8.9 所示电路为正交正弦波振荡电路，它可产生频率相同的正弦信号和余弦信号。已知稳压管的稳定电压 $\pm U_Z = \pm 6V$ ， $R_1 = R_2 = R_3 = R_4 = R_5 = R$ ， $C_1 = C_2 = C$ 。

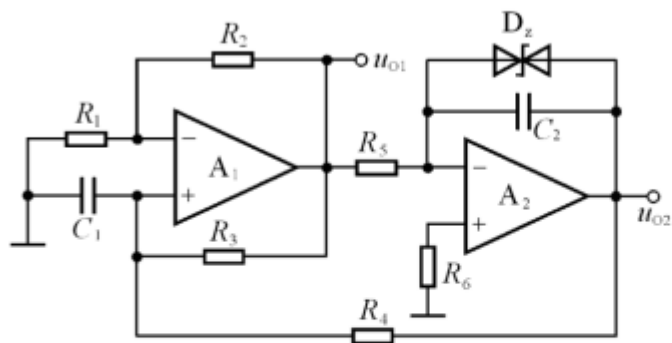


图 P8.9

- (1) 试分析电路为什么能够满足产生正弦波振荡的条件；
- (2) 求出电路的振荡频率；
- (3) 画出 u_{O1} 和 u_{O2} 的波形图，要求表示出它们的相位关系，并分别求出它们的峰值。

8.14 please figure out the voltage comparison relationship and draw out the curve between the input and output voltages.

8.14 试分别求解图 P8.14 所示各电路的电压传输特性。

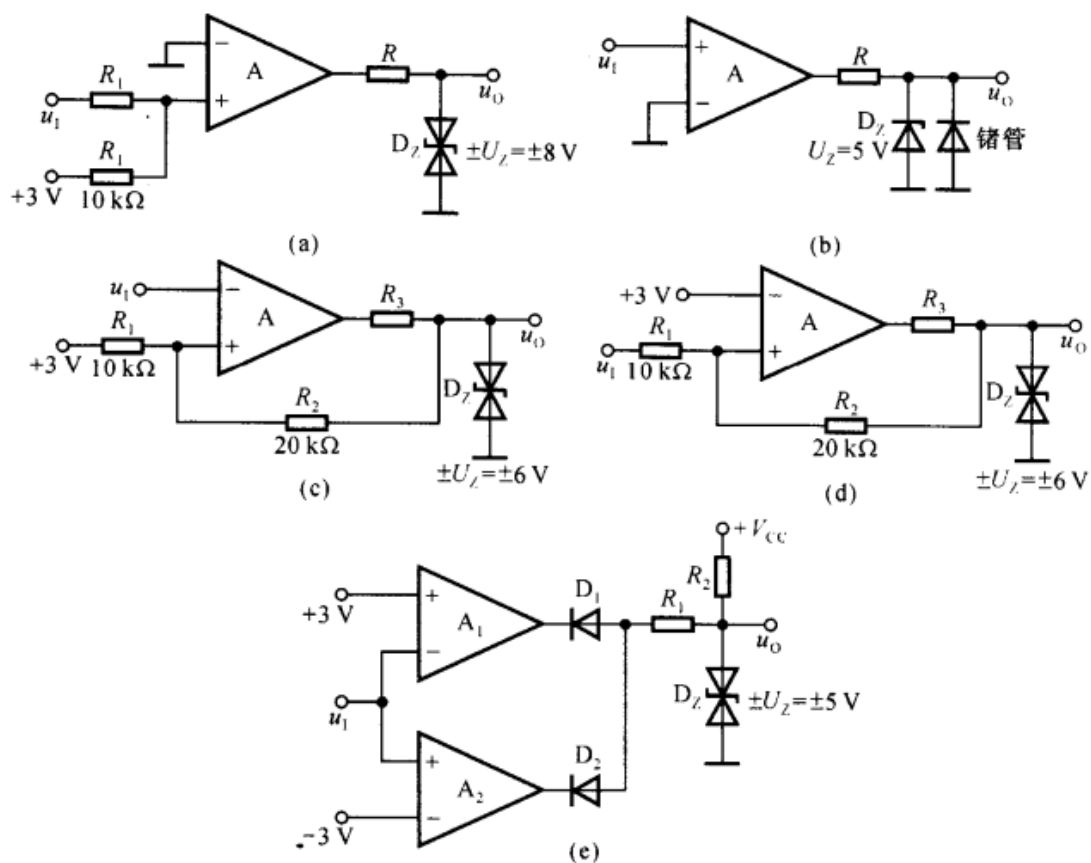


图 P8.14

8.15 The voltage transmission characteristics of the three voltage comparators are known as (a) (b) (c), and their input voltage waveforms are shown in figure (d), draw the waveforms of u_{o1} , u_{o2} and u_{o3} .

8.15 已知三个电压比较器的电压传输特性分别如图 P8.15 (a)、(b)、(c)所示，它们的输入电压波形均如图(d)所示，试画出 u_{o1} 、 u_{o2} 和 u_{o3} 的波形。

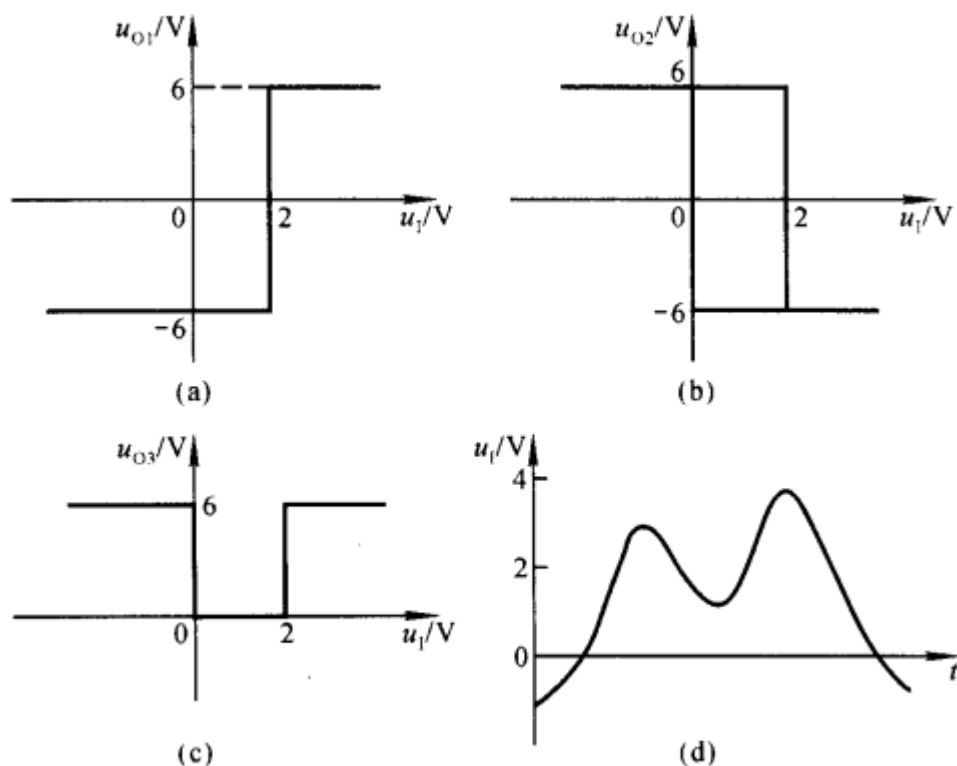


图 P8.15

8.18 In the circuit shown in Figure P8.18, it is known that $R_1=10\text{k}\Omega$, $R_2=20\text{k}\Omega$, $C=0.01\text{ }\mu\text{F}$, the maximum output voltage amplitude of the integrated operational amplifier is $\pm 12\text{V}$, and the dynamic resistance of the diode is negligible.

(1) Find the oscillation period of the circuit;

(2) Draw the waveforms of u_o and u_c .

8.18 在图 P8.18 所示电路中，已知 $R_1 = 10\text{ k}\Omega$ ， $R_2 = 20\text{ k}\Omega$ ， $C = 0.01\text{ }\mu\text{F}$ ，集成运放的最大输出电压幅值为 $\pm 12\text{ V}$ ，二极管的动态电阻可忽略不计。

(1) 求出电路的振荡周期；

(2) 画出 u_o 和 u_c 的波形。

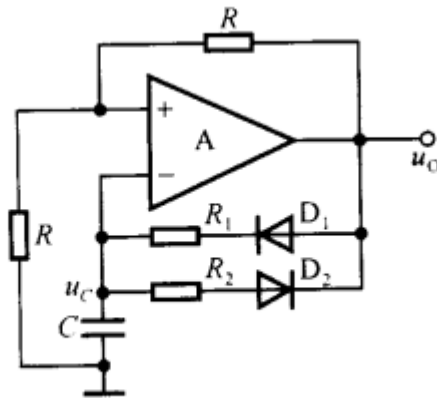


图 P8.18

8.19 As shown in the picture, the square wave generator is designed by a student . find out the three errors in the diagram and to correct it.

8.19 图 P8.19 所示电路为某同学所接的方波发生电路，试找出图中的三个错误，并改正。

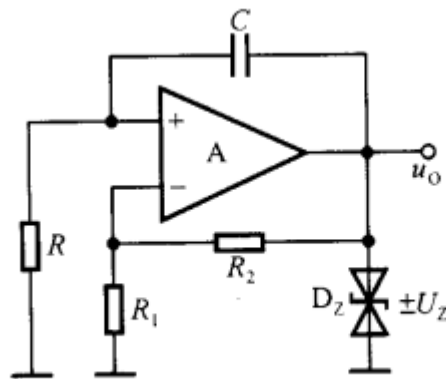


图 P8.19

8.20 The waveform generation circuit is shown as shown in Figure 8.20

If the oscillation period is T and the time of $U_{O1}=U_Z$ is T_1 in one period, the duty ratio is T_1/T ;

$R_{w1} \ll R_{w2}$;

When one parameters of the circuit changes, the rest of the parameters are unchangedl.

Choose “ increase, no change, decrease” and fill in the empty space

(1) When R_1 increases, the duty ratio of U_{O1} will _____; oscillation frequency will _____;
The amplitude of U_{O2} will _____;

(2) If the sliding end of the R_{w1} moves upwards, the duty ratio of U_{O1} will _____; oscillation frequency will _____; The amplitude of U_{O2} will _____;

(3) If the sliding end of the R_{w2} moves upwards, the duty ratio of U_{o1} will _____; oscillation frequency will _____; The amplitude of U_{o2} will _____;

8.20 波形发生电路如图 P8.20 所示, 设振荡周期为 T , 在一个周期内 $u_{o1} = U_Z$ 的时间为 T_1 , 则占空比为 T_1/T ; $R_{w1} \ll R_{w2}$; 在电路某一参数变化时, 其余参数不变。选择①增大、②不变或③减小填入空内:

当 R_1 增大时, u_{o1} 的占空比将 _____, 振荡频率将 _____, u_{o2} 的幅值将 _____; 若 R_{w1} 的滑动端向上移动, 则 u_{o1} 的占空比将 _____, 振荡频率将 _____, u_{o2} 的幅值将 _____; 若 R_{w2} 的滑动端向上移动, 则 u_{o1} 的占空比将 _____, 振荡频率将 _____, u_{o2} 的幅值将 _____。

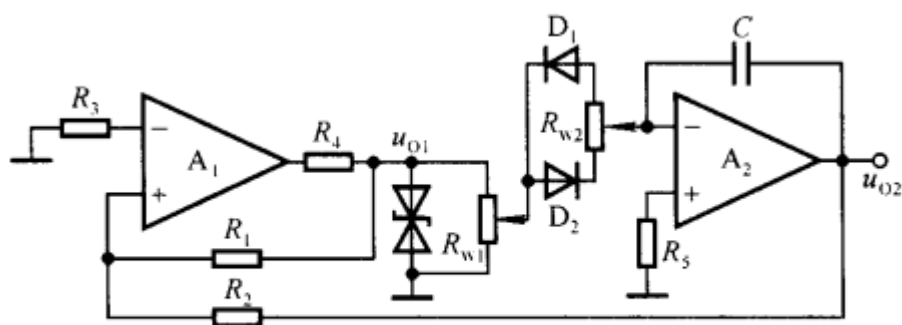


图 P8.20

8.21 In the circuit shown in Figure 8.20, the sliding end of the R_{w1} is at the top end, draw the waveform of the U_{o1} and U_{o2} when the R_{w2} 's sliding end at the top and at the bottom.

8.21 在图 P8.20 所示电路中, 已知 R_{w1} 的滑动端在最上端, 试分别定性画出 R_{w2} 的滑动端在最上端和在最下端时 u_{o1} 和 u_{o2} 的波形。